
JOB DESCRIPTION



VLSI Technology

Very Large Scale Integration

Semiconductor Division-2, Nagarbhavi, Bengaluru

www.vlsiind.in

Semiconductor R&D wing

AI/ML Engineer

1. Company Profile

VLSI Technology is a core VLSI Design Company with nearly two decades of expertise in core VLSI Design, Semiconductors, and SoC chip Design, with a strong focus on R&D in Product Design and Development. We cater to the Indian market by offering a diverse range of innovative products in the Healthcare and Aerospace Engineering domains.

2. Our Areas of Expertise

VLSI Design, Semiconductors, SoC6, SoC7, 760nm Technology, Chipset AI Studio, Chiplet 3D Integration, and AI-Embedded Design Hardware and Software.

3. Job Title: AI/ML Engineer

Job Entry Type: Research-Based Intern

4. Target Students:

Bachelor's and Master's degree in Computer Engineering, Information Technology, AIML, IoT, Data science, Cloud Computing, Cyber Security

5. CTC: 6 LPA to 12 LPA

6. Duration of Internship: 4 to 6 months or till document verification.

7. Prerequisites (Qualifications & Skills)

To qualify for this role in the **AI / ML Engineer** domain, you need a strong foundation in **mathematics, machine learning fundamentals, and practical model development.**

Required Knowledge:

- **Strong understanding of mathematics for machine learning**, including linear algebra, probability, statistics, and basic optimization techniques.
- **Proficiency in Python for AI/ML development**, with working knowledge of libraries such as **NumPy, Pandas, Scikit-learn, PyTorch, and TensorFlow.**
- **Solid grasp of machine learning concepts**, including supervised and unsupervised learning, feature engineering, model evaluation, and overfitting/underfitting.
- **Understanding of deep learning fundamentals**, including neural networks, CNNs, RNNs/Transformers, loss functions, and backpropagation.
- **Familiarity with data preprocessing and pipeline development**, including data cleaning, normalization, augmentation, and train/validation/test workflows.
- **Experience or exposure to deploying ML models**, including model serialization, inference pipelines, and integration with APIs or production systems.
- **Basic awareness of performance optimization and scalability**, including hardware acceleration (CPU/GPU), training efficiency, and monitoring model performance in production.

8. Location:

Usually onsite/CoE established at College/University (requires lab access for usage of components/boards), any company division across in INDIA

9. Experience Level:0–2years (Entry-Level/Junior)

10. Job Role (Responsibilities):

As an **AI / ML Engineer**, you will support the end-to-end lifecycle of machine learning solutions, from data preparation and model development to deployment, monitoring, and continuous improvement.

Data Preparation & Feature Engineering: Collecting, cleaning, and transforming large datasets into features suitable for training and evaluating machine learning models.

Model Development & Training: Designing, implementing, and training machine learning and deep learning models using frameworks such as TensorFlow, PyTorch, or Scikit-learn.

Algorithm Evaluation & Validation: Testing and validating models using appropriate metrics to ensure accuracy, robustness, and generalization.

Model Deployment & Integration: Deploying trained models into production environments and integrating them with applications through APIs or inference pipelines.

Optimization & Performance Tuning: Optimizing models for performance, scalability, and resource efficiency, including latency, memory usage, and hardware utilization.

Monitoring & Debugging: Monitoring deployed models for performance drift, data quality issues, and system failures, and troubleshooting model or pipeline issues.

Documentation: Creating and maintaining technical documentation for model architectures, training processes, evaluation metrics, and usage guidelines for stakeholders and end users.

11. Onboarding

Onboarding in a semiconductor firm is often more rigorous than in standard IT roles due to the high cost of chip manufacturing.

Technical Induction/Free of Cost Training: A 4-6 months deep-dive into the company's proprietary AI-Embedded Architecture and internal software development toolchains and engineering workflows under the **PIE(Performance in Evaluation) model**, during which three assessments will be given out of which it is mandatory to clear two assessments.

NDA & IP Protection: Strict legal agreements regarding Intellectual Property (IP); you will be working on designs that are often 2–3 years away from hitting the Research market .

Mentorship Period: You will typically be assigned a "Buddy" or "Mentor" (Senior Design Engineer) for the first 6 months to oversee your code commits.

Probationary Project: Completion of a "Pilot Project" such as developing a driver for a new sensor interface to demonstrate competency.

The Industry Reality: In the VLSI domain, a mistake in your AI-Embedded code can sometimes be fixed with a software patch, but a mistake in how that code interacts with the hardware logic might require a multi-million dollar "re-spin" of the chip. This is why accuracy and attention to detail are the most valued soft skills in this JD.

Note:

Job roles and responsibilities are not limited to above specifications; the roles vary as per the company requirement and demand of client. The candidate should be flexible to adapt as per company requirement anywhere anytime. Shortlisted and selected students are required to sign a Consent Form and a Non-Disclosure Agreement (NDA) before proceeding.

The stated CTC applies only to candidates who clear all eligibility rounds and criteria in Annexure-I. If you are a referred candidate or applied via off-campus recruitment process or through internship model, you will be entering VLSI Technology as an intern under PIE model (Performance In Evaluation).

12. A. Interview Process

Round 1: Written Test (Aptitude & Fundamentals)

Mode: Offline

Duration: 30 minutes

Objective:

To evaluate the candidate's core fundamentals, analytical ability, and basic technical knowledge.

Round 2: Open book Coding Test (Hands-on Programming)

Mode: Offline

Requirement: Candidates must carry their own laptop

Duration: 45 minutes

Objective:

To assess practical coding skills, problem-solving approach, and code quality.

Round 3: Face-to-Face Technical + HR Interview

Mode: In-person

Duration: 30–45 minutes

Objective:

To evaluate technical depth, communication skills, attitude, and cultural fit, domain assignment.

